

Frozen-Encoder Baselines

Linear probes and image-derived prototypes on DTD and FGVC-Aircraft

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Stage 1 is the ground that the rest of the project stands on. Before we put a flow-matching layer anywhere near a classifier, we needed to know how well two frozen encoders classify two datasets when all they get is a linear probe or a class-mean prototype, and we needed those numbers to be solid enough to compare against in Stages 2 and 3. This report follows our five notebooks in order. For each one we describe what it did, show the figures that best explain the data and the results, and give the reasoning behind the choices we made on the way. Together the sections cover everything we set out to measure: the accuracy table, accuracy against training-set size, training curves, confusion matrices, and feature visualisations.

Datasets	DTD (47 texture classes, official partition 1) and FGVC-Aircraft (100 variant-level classes)
Frozen encoders	ResNet-18 (512-d, both datasets); DINOv2 ViT-S/14 (384-d, Aircraft only)
Baselines	A trained linear probe, $s = Wz + b$, and image-derived class prototypes (no training)
Training sizes	$K \in \{5, 10, \text{full}\}$. 5- and 10-shot use 3 balanced random subsets; full uses the complete training split with 3 seeds
Metric	Top-1 accuracy on the complete, untouched official test split, every time

1 Data setup and protocol verification NOTEBOOK 1

We did not extract a single feature until the data itself had been checked. Notebook 1 loads the official splits of both datasets and runs twenty automated checks, ten per dataset: the class count and partition level, the split sizes against the published counts, pairwise disjointness of train, validation and test, the presence on disk of every referenced file, and a floor of at least ten training images per class so that 10-shot sampling is always possible. All twenty pass before Notebook 2 ever runs an encoder.

Check category	DTD	FGVC-Aircraft
Class count and partition/level correct	PASS	PASS
Split sizes match official counts	PASS	PASS
Train / val / test pairwise disjoint	PASS	PASS
Every referenced file present on disk	PASS	PASS
≥ 10 training images per class (10-shot floor)	PASS	PASS

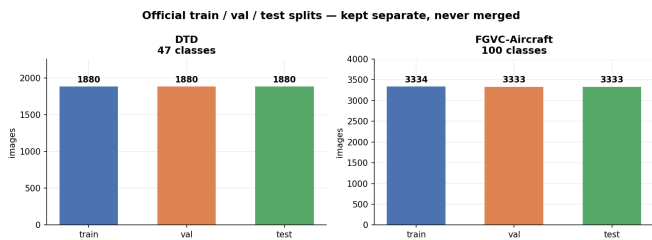


Figure 1. Official split sizes. Both datasets happen to divide into near-equal thirds (DTD 1880/1880/1880, Aircraft 3334/3333/3333). That proportion comes from the dataset authors, not from us; we keep the three splits separate and never merge them.

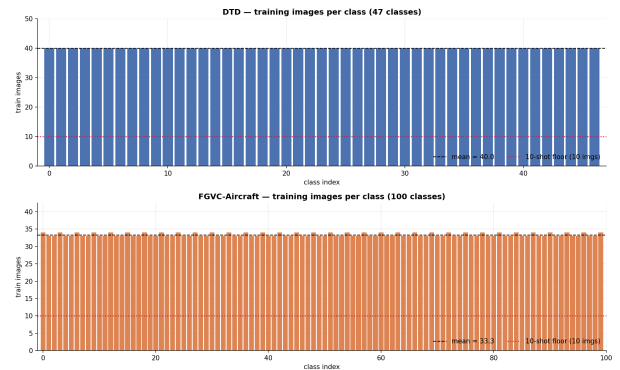


Figure 2. Training images per class. DTD is flat at 40 per class; Aircraft averages about 33. Every class sits comfortably above the red 10-shot floor.

We also looked at what the images physically are, because that turned out to matter for preprocessing. Aircraft photographs cluster tightly at an aspect ratio of about 1.47, a wide landscape frame (Figure 3). A standard 256-resize followed by a 224 centre crop therefore keeps only around 59% of an aircraft image’s width, whereas DTD textures, which are orientation-uniform, lose far less on average (Figure 4).

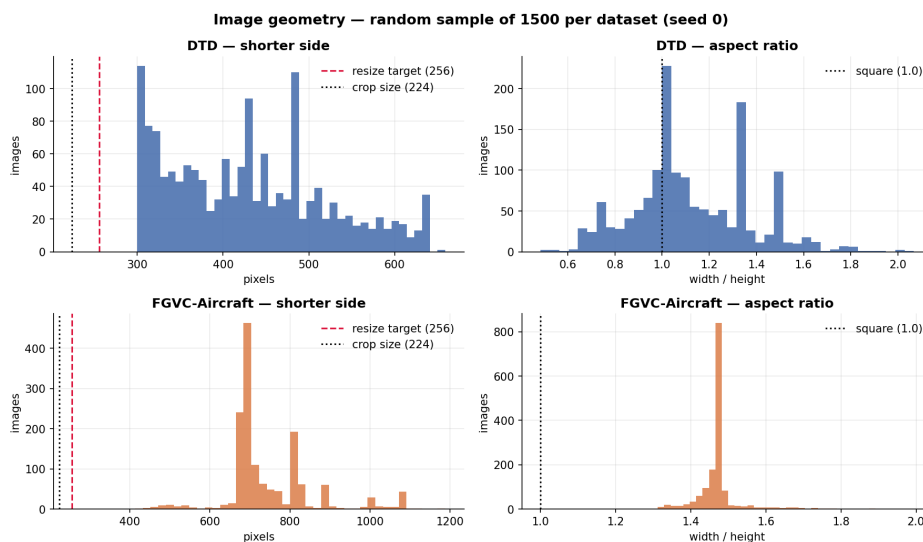


Figure 3. Shorter-side and aspect-ratio distributions on a random sample of 1500 images per dataset. The Aircraft aspect-ratio spike at about 1.47 is what makes the centre crop expensive for that dataset.

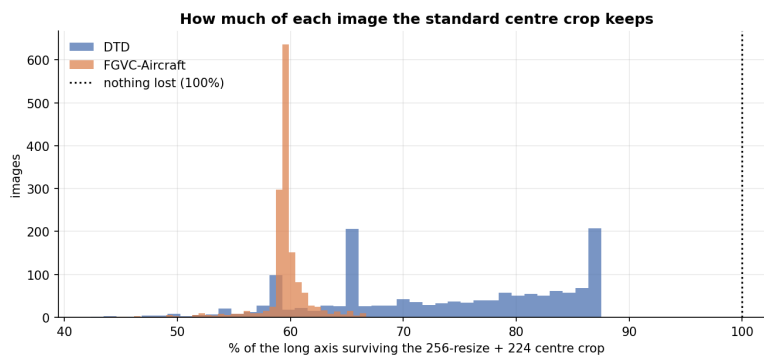


Figure 4. Fraction of the long axis that survives the 256-resize + 224 centre crop. Aircraft images keep only about 59% of their width; DTD keeps much more.

PREPROCESSING DECISION 1 · THE CENTRE CROP STAYS

We use the pretrained checkpoint’s own preprocessing (resize 256, centre-crop 224) unchanged for both datasets, even knowing it discards roughly 41% of each Aircraft image’s width. Deviating from the published preprocessing would feed the encoder an input distribution it was not pretrained on, and it would break comparability both with the frozen encoder itself and with Stages 2 and 3, which reuse the same cached features. We record this as a known, quantified limitation rather than something we changed.

There is a second, unrelated quirk in the Aircraft data. Every FGVC-Aircraft image carries a 20-pixel copyright banner (for example COPYRIGHT MICK BAJCAR) burned directly into the bottom rows of pixels (Figure 5).



Figure 5. Three Aircraft images and a zoom on their bottom 30 px. The red line marks the crop we apply; the banner text is a printed artifact of the source images, not part of the photograph.

PREPROCESSING DECISION 2 · THE BANNER GOES

This is a different decision from the centre crop and we keep it separate. The banner is not part of the checkpoint’s preprocessing; it is a printed artifact on the source images, unrelated to what the pretrained model expects. Cropping it (done in Notebook 2) is free, strictly more correct, since the encoder then sees a photograph rather than a photograph plus a copyright notice, and it does not touch comparability with the checkpoint. Notebook 2 then measures directly whether the crop changes accuracy. It does not, within noise, so we treat it as hygiene and do not claim it as an improvement.

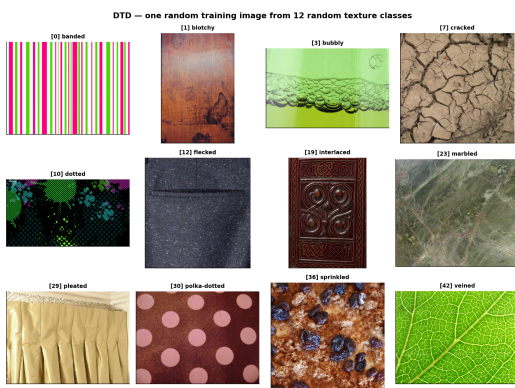


Figure 6. One random training image from each of twelve DTD classes: a cheap check that catches a corrupted extraction or a label/image misalignment that accuracy numbers alone would hide.



Figure 7. Training images used at each K . The full set is $8\times$ the 5-shot set on DTD and $6.7\times$ on Aircraft, enough spread to keep the three settings clearly distinct.

2 Feature extraction NOTEBOOK 2

Notebook 2 runs both frozen encoders once over every image and caches the result. Nothing touches an image again after this point; every later notebook, and Stages 2 and 3, work from the same cache. The run produced nine cache files covering 25,640 encoded images, each of which we verified against the source labels after reloading from disk.

Encoder	Checkpoint	Feature dim	Params	Trainable
ResNet-18	torchvision IMAGENET1K_V1	512	11.18M	0
DINOv2 ViT-S/14	facebookresearch/dinov2	384	22.06M	0

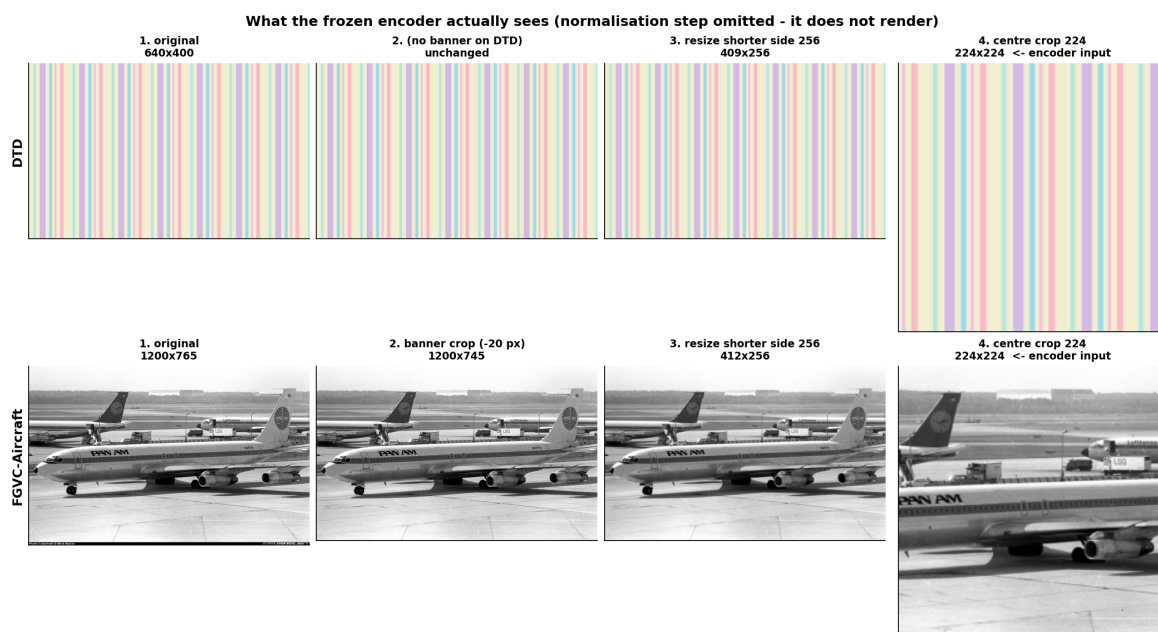


Figure 8. One real image from each dataset traced through every stage: original → banner crop (Aircraft only) → resize 256 → centre-crop 224. The rightmost panel is exactly what the encoder receives.

Before training any classifier we used the cached features on their own to run a few diagnostics. In hindsight these predicted how the rest of the stage would go. The most telling one is the cosine-similarity margin between a test feature and its own class centroid versus the other class centroids (Figure 9): 0.172 for ResNet-18 on DTD, only 0.041 for ResNet-18 on Aircraft, and 0.420 for DINOv2 on Aircraft. Training-free nearest-centroid accuracy follows the same order (67.7%, 40.4%, 44.6%). With no classifier in sight, the features already say that ResNet-18 will struggle on Aircraft and DINOv2 will not, which is exactly what Notebooks 3 and 4 go on to find.

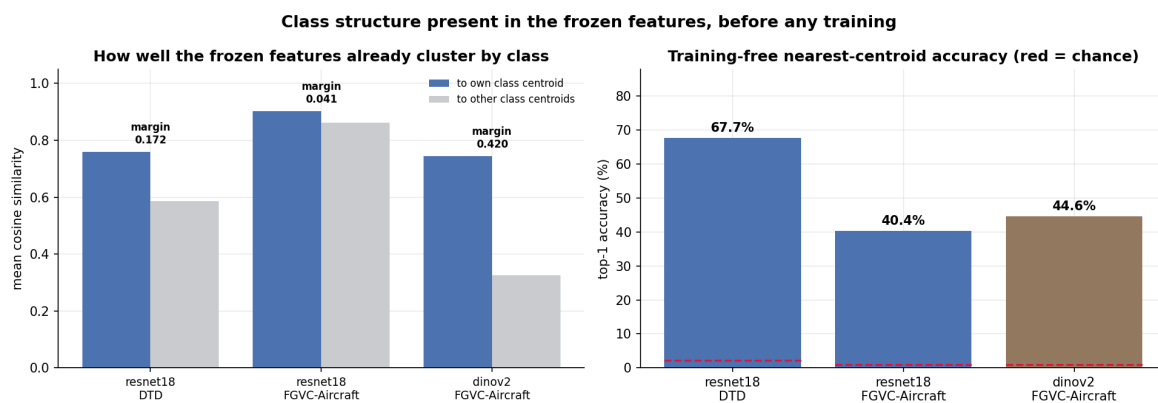


Figure 9. Class structure present in the frozen features before any training. Left: mean cosine similarity to own versus other class centroids. Right: nearest-centroid top-1 accuracy on the test split (red line marks chance).

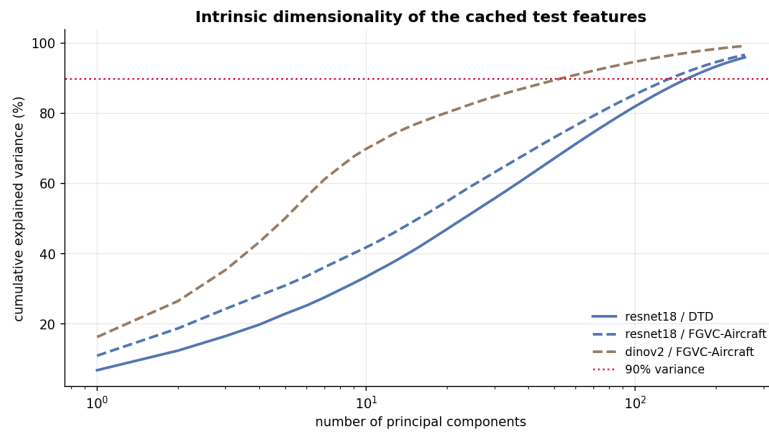


Figure 10. Cumulative explained variance of the cached test features. No encoder’s feature space is degenerate (DINOv2 needs 104 of its 384 dimensions to reach 95%), so weak accuracy later cannot be blamed on a collapsed representation.

The feature geometry also has practical consequences later on. Feature norms sit around 24 to 50 depending on the encoder and are never near 1, ResNet-18 features are non-negative (they come after a ReLU) with 7.3% exact zeros on DTD, and DINOv2 features are signed and dense (Figure 11). Since prototypes are unit-norm by construction, any joint plot of features and prototypes has to L2-normalise first; Section 5 shows what goes wrong when it does not.

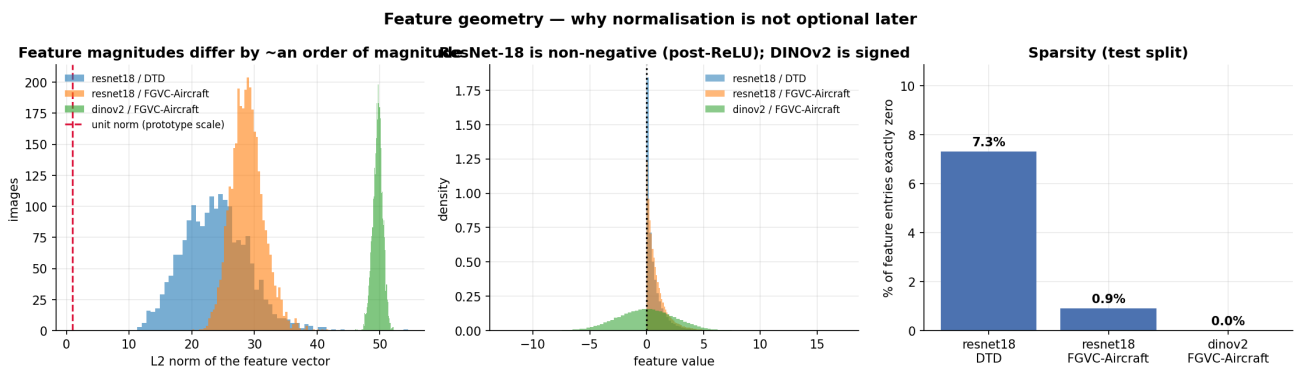


Figure 11. Feature magnitudes, value distributions and sparsity on the test split. Norms differ by roughly an order of magnitude between encoders.

Finally, we closed the loop on the banner crop from Section 1 with a direct ablation (Figure 12). Cropped and uncropped features of the same image have a mean cosine similarity of 0.982, and on 900 test images nearest-centroid top-1 moves from 32.00% with the banner kept to 31.22% with it cropped. That 0.78 pp change is inside the ± 1.55 pp noise margin, which is why we call the crop hygiene rather than an improvement.

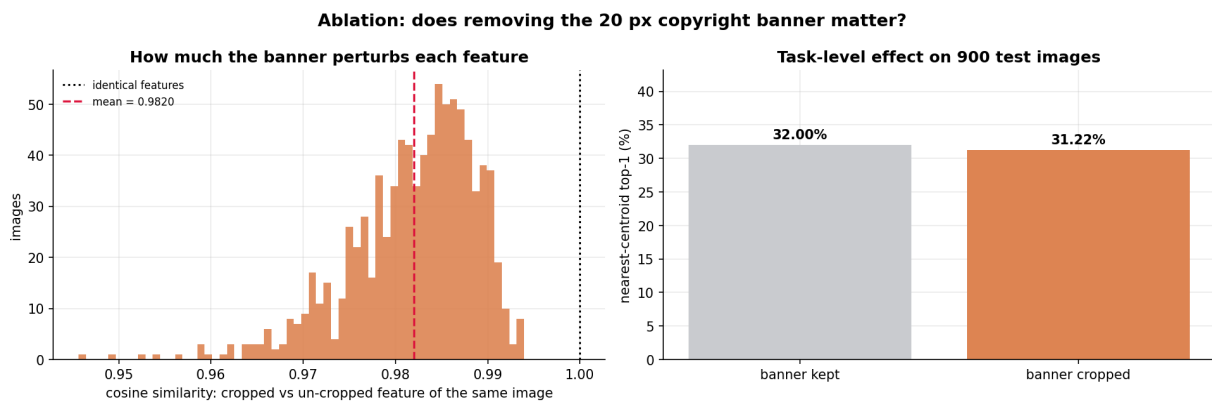


Figure 12. Banner ablation. Left: per-image cosine similarity between cropped and uncropped features. Right: task-level effect on 900 test images.

3 Linear probe NOTEBOOK 3 · FIRST BASELINE

We trained $s = Wz + b$ on the cached features: 27 runs in total, across three dataset/encoder combinations, three K settings and three seeds.

Dataset / Encoder	5-shot	10-shot	full
DTD / ResNet-18	44.86 $\pm$ 1.83	52.04 $\pm$ 1.16	62.82 $\pm$ 0.42
Aircraft / ResNet-18	20.76 $\pm$ 0.26	27.87 $\pm$ 1.33	38.15 $\pm$ 0.29
Aircraft / DINOv2	38.19 $\pm$ 2.49	51.54 $\pm$ 1.70	67.55 $\pm$ 0.28

Table 1. Linear-probe top-1 test accuracy (%), mean $\pm$ std over 3 seeds.

The training configuration (AdamW, learning rate 1e-3, weight decay 1e-4, batch size 64, up to 200 epochs, checkpoint chosen on highest validation accuracy) is a standard one, and we kept it unmodified. We would have adjusted it had it misbehaved, but it did not: no run diverged, none produced NaN or Inf, and every run selected a real checkpoint, as the stability and overfitting figures below show.

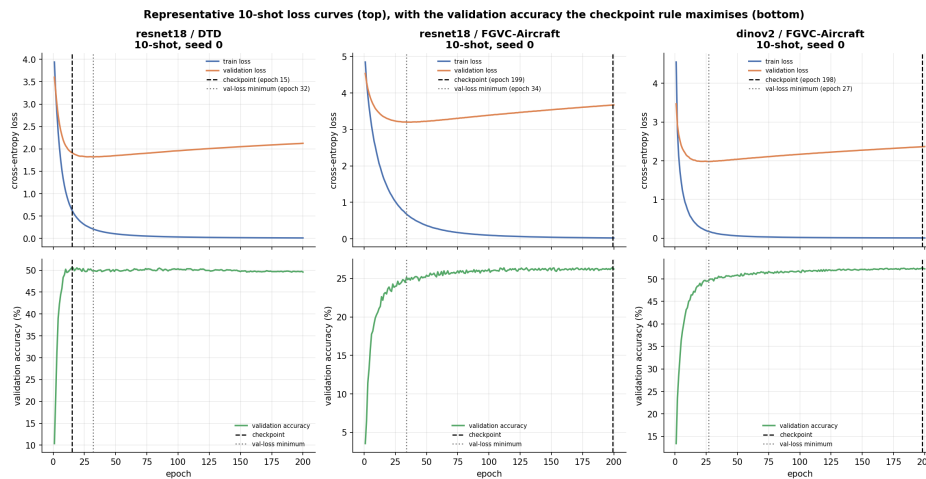


Figure 13. Representative 10-shot training curves (seed 0). Top: train and validation cross-entropy with the selected checkpoint and the validation-loss minimum marked. Bottom: the validation accuracy the checkpoint rule maximises.

Training is stable in every setting. The one pattern worth explaining is on Aircraft, where validation accuracy keeps climbing long after the validation loss bottoms out around epoch 35 and starts to rise. This is expected behaviour rather than a fault: cross-entropy penalises over-confident mistakes, so the loss can grow while the argmax keeps getting better. Figure 14 shows the same shape holds across all three seeds and all three K settings, not just the one curve we picked as representative.

Where the checkpoint lands tells the two datasets apart cleanly (Figure 15). DTD peaks around epoch 18–32 and spends the rest of the budget overfitting, while Aircraft is still improving past epoch 150, a difference that tracks the separability gap we measured in Notebook 2. Because the late Aircraft checkpoints looked unusual, we tested them directly instead of leaving it as a guess: tripling the epoch budget changed accuracy by at most 0.21 pp.

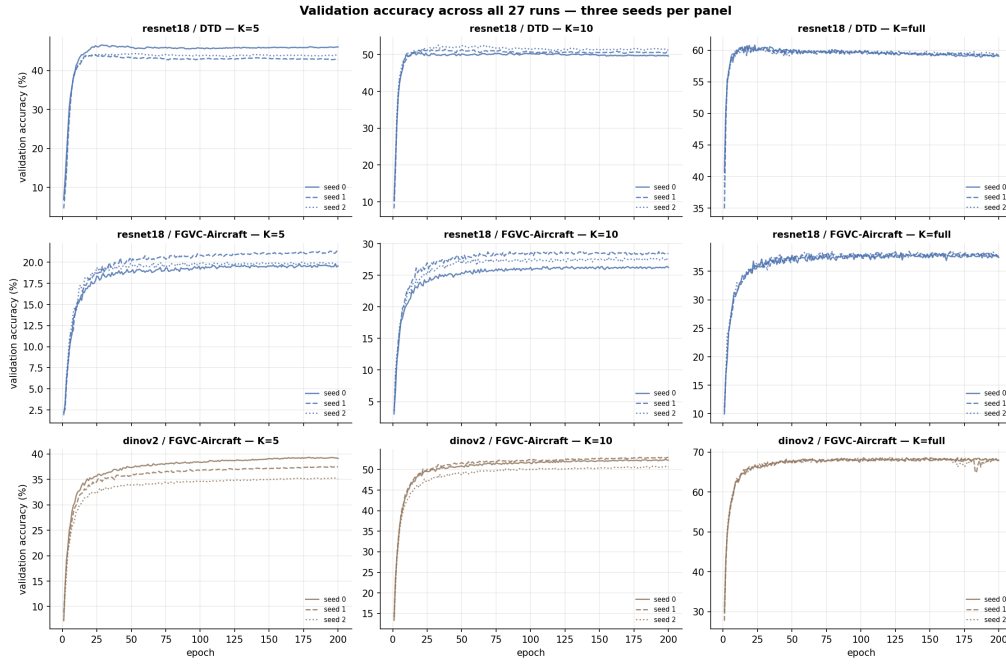


Figure 14. Validation accuracy across all 27 runs, three seeds per panel.

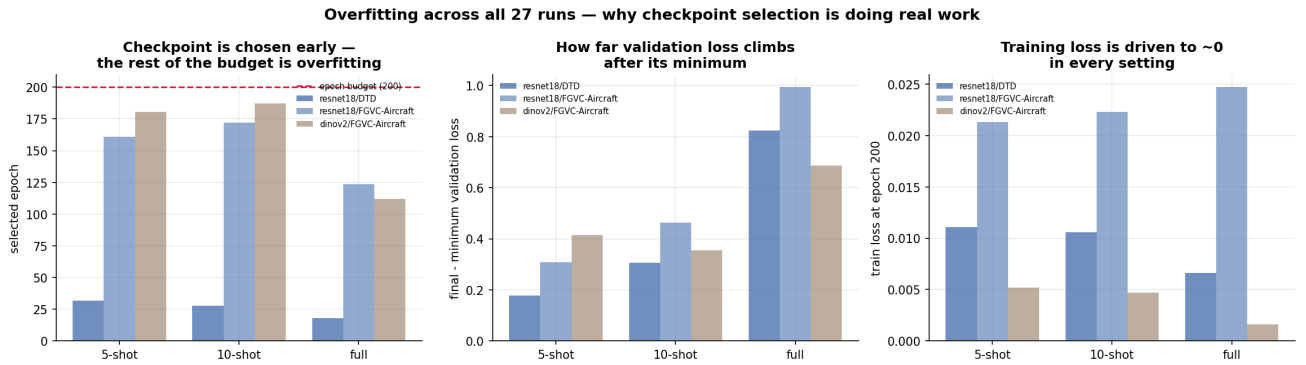


Figure 15. Selected epoch, how far validation loss climbs after its minimum, and final training loss across all 27 runs. Checkpoint selection is doing real work on DTD; training loss is driven to about 0 everywhere.

As a last check on the probe we compared two independent implementations of the same protocol: our first version of the pipeline and the final one, which re-extracts the features and uses a rewritten training loop (Figure 16). On DTD, which receives no banner crop and so isolates the reimplementation itself, the two agree within 0.26 pp. We take that as strong evidence that both implementations are correct.

4 Image-derived prototypes NOTEBOOK 4 · SECOND BASELINE

The second baseline needs no training at all. For each class we average its L2-normalised training features into a single prototype, and a test image is assigned to the prototype with the highest cosine similarity.

Dataset / Encoder	5-shot	10-shot	full
DTD / ResNet-18	46.10 $\pm$ 1.89	51.67 $\pm$ 1.33	58.83 (1 run)
Aircraft / ResNet-18	16.46 $\pm$ 0.39	20.19 $\pm$ 1.14	25.47 (1 run)
Aircraft / DINOv2	23.79 $\pm$ 0.47	27.74 $\pm$ 0.98	34.26 (1 run)

Table 2. Prototype top-1 test accuracy (%). The full-data setting is deterministic, so it has a single run.

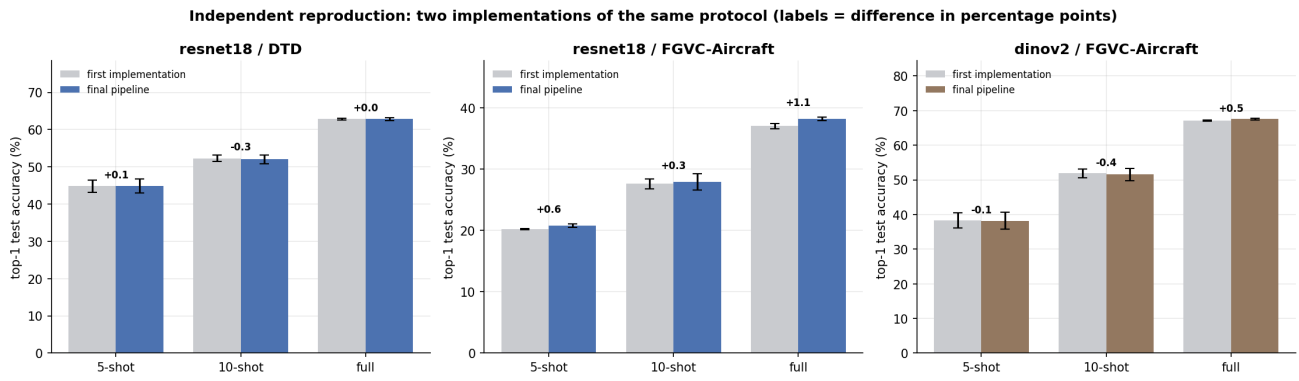


Figure 16. Independent reproduction. Labels give the difference in percentage points between the final pipeline and our first implementation.

WHY PROTOTYPES RATHER THAN ZERO-SHOT CLIP

Image-derived prototypes reuse the exact frozen features already cached in Notebook 2 instead of introducing a fourth pretrained model, and they share the same k-shot subsets as the linear probe (same seeds, same images). That second point is what turns the comparison in Section 5 into a statement about classifier design rather than an artifact of different training data.

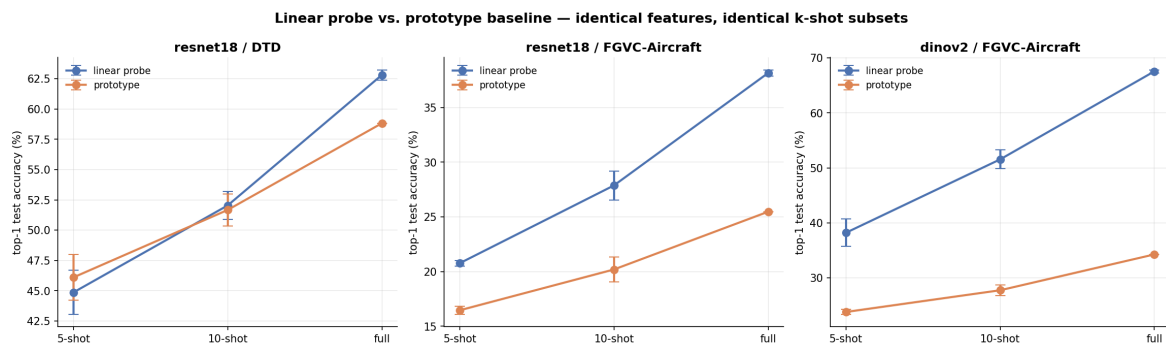


Figure 17. Linear probe versus prototypes on identical features and identical k-shot subsets. The two are close on DTD (prototypes even edge ahead at 5-shot); on Aircraft the probe pulls away as K grows, by up to 33 pp at full data.

The reason is not subtle. Averaging a class into one vector destroys the fine-grained detail that separates one aircraft variant from another, and a trained boundary can still use that detail. The aggregate numbers also hide how uneven the prototype classifier is per class. On ResNet-18/Aircraft, 85 of the 100 classes score below 50% own-prototype accuracy even though the overall figure is 25.4%; a handful of distinctive classes carries the average (Figure 18). Figure 19 puts names to the largest gaps, and Figure 20 shows that even when the prototype classifier is right, it is often only barely right.

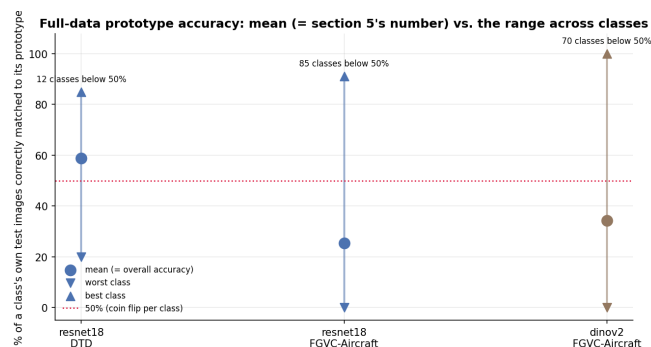


Figure 18. Full-data prototype accuracy: the overall mean against the best and worst class. The red line is the 50% coin-flip level per class.

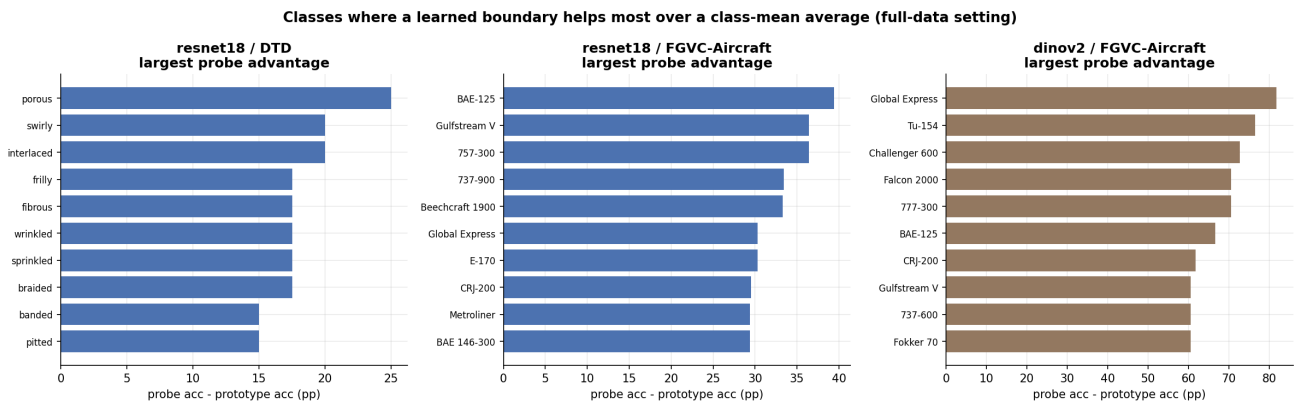


Figure 19. The ten classes per setting where a learned boundary helps most over a class-mean average. On DINOv2/Aircraft, Global Express, Tu-154 and Challenger 600 each lose 70–82 pp to the linear probe.

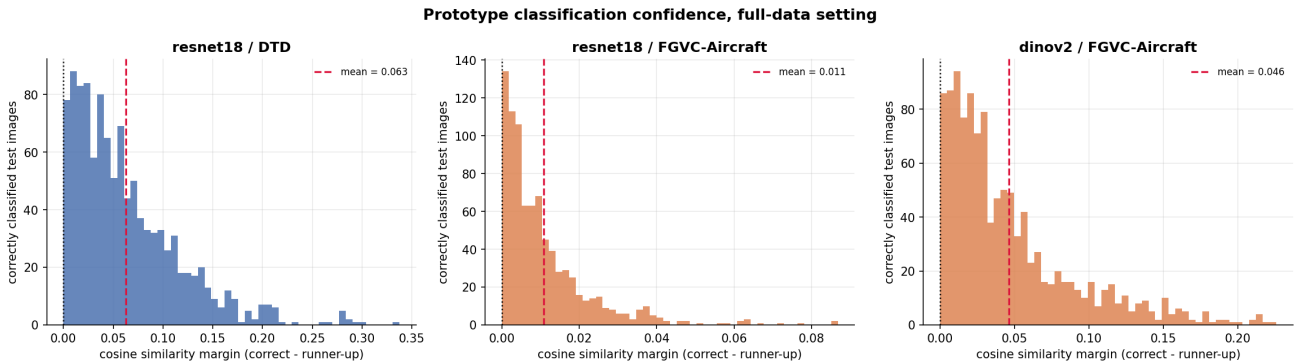


Figure 20. Confidence margins on correct prototype predictions (winning similarity minus runner-up). The thin Aircraft margins are consistent with the low separability measured in Notebook 2.

5 Analysis NOTEBOOK 5

Notebook 5 assembles the headline results from the saved outputs of Notebooks 2 to 4; nothing is retrained or re-extracted. The complete accuracy table comes first.

Dataset / Encoder	K	Linear probe	Prototype
DTD / ResNet-18	5	44.86 ± 1.83	46.10 ± 1.89
	10	52.04 ± 1.16	51.67 ± 1.33
	full	62.82 ± 0.42	58.83 (1 run)
Aircraft / ResNet-18	5	20.76 ± 0.26	16.46 ± 0.39
	10	27.87 ± 1.33	20.19 ± 1.14
	full	38.15 ± 0.29	25.47 (1 run)
Aircraft / DINOv2	5	38.19 ± 2.49	23.79 ± 0.47
	10	51.54 ± 1.70	27.74 ± 0.98
	full	67.55 ± 0.28	34.26 (1 run)

Table 3. Top-1 test accuracy (%) for every dataset, encoder, K and baseline. Accuracy against training-set size is the same data plotted in Figure 17: both baselines improve monotonically with K , and the probe's lead widens fastest on Aircraft.

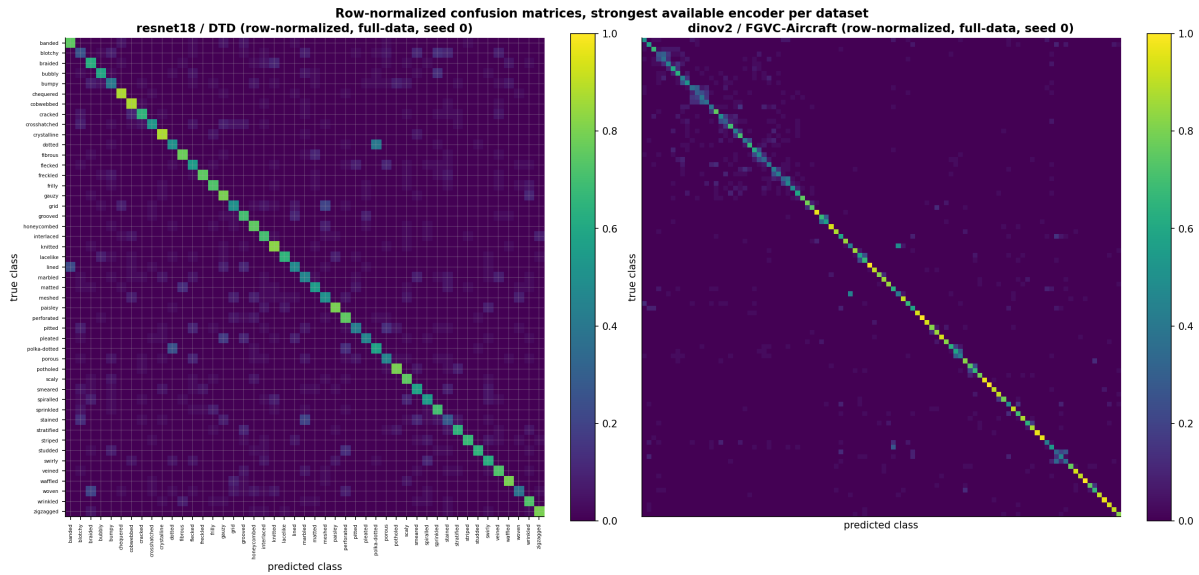


Figure 21. Row-normalised confusion matrices for the strongest encoder on each dataset (full data, seed 0). DTD’s 47 classes are labelled directly; Aircraft’s 100 are not, purely for readability.

Dataset	Top confusion	Rate
DTD	dotted → polka-dotted	40.0%
DTD	polka-dotted → dotted	25.0%
DTD	lined → banded	22.5%
Aircraft	C-47 → DC-3	48.5%
Aircraft	DC-3 → C-47	44.1%
Aircraft	A340-200 → A340-300	38.2%

Table 4. The largest off-diagonal entries in Figure 21.

The confusions are the kind we would hope to see from a sensible model. On DTD they sit among genuinely related pattern concepts. On Aircraft they sit inside manufacturer families: the C-47 and DC-3 are both classic twin-prop transports that look nearly identical, and the A340-200 and A340-300 are the same airframe built to two lengths. This is the same story the feature-space analysis tells from the other direction.

FEATURE VISUALISATION CHOICES

We use PCA rather than t-SNE. It is deterministic, with no perplexity or seed to justify, and fitting the projection jointly to the image features and the prototypes has an unambiguous meaning for a linear method. The nine classes shown per dataset were chosen by an evenly spaced stride through the sorted class list, not by hand, so the subset is diverse without being cherry-picked.

Figure 22 shows why the features have to be normalised before this projection. Prototypes are unit-norm by construction but raw features are not (norms of 24 to 50, from Notebook 2), so projecting the two together without normalising sends every prototype into one corner of the plot. After L2-normalising the features, each prototype sits inside its own class’s cloud, which is the plot that actually answers the question.

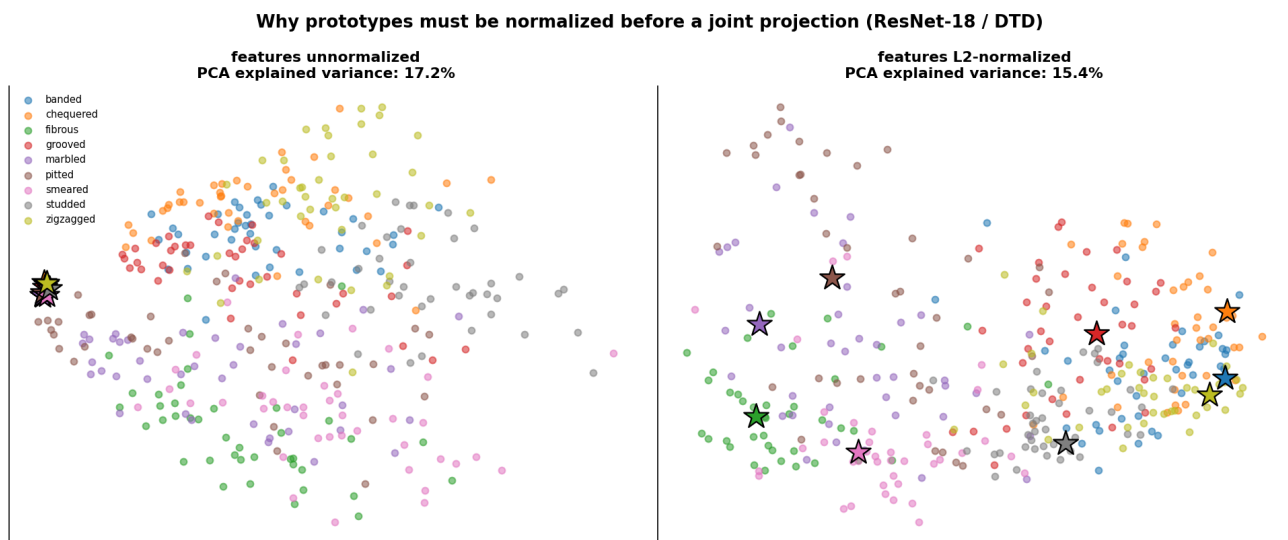


Figure 22. ResNet-18 / DTD test features with class prototypes (stars), nine classes. Left: features left unnormalised, prototypes collapse into one corner. Right: after L2-normalising, each star lands in its own class.

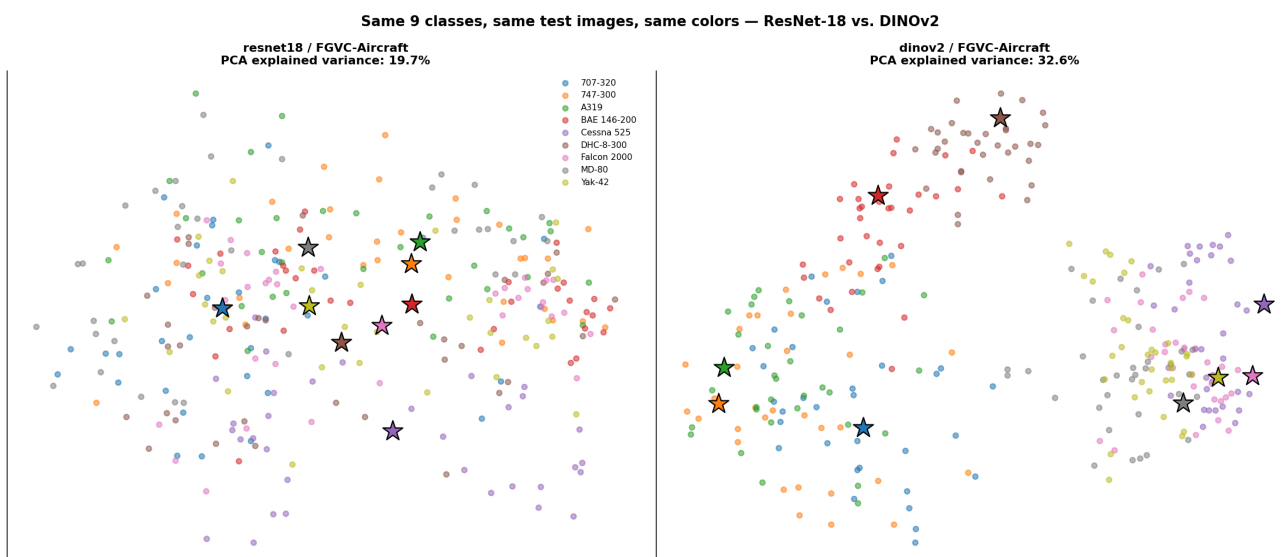


Figure 23. The same nine Aircraft classes, the same test images and the same colours under ResNet-18 (left) and DINOv2 (right). DINOv2's classes separate into visibly tighter clusters, and its projection explains 32.6% of the variance against 19.7%. We read this qualitatively; a 2-D projection is not a performance measurement.

6 What we take from Stage 1

Both baselines are implemented and evaluated under one protocol, with three K settings, three seeds, and the full official test split every time. The results hang together: the separability we measured on the raw features in Notebook 2 predicts the accuracies of Notebooks 3 and 4, the confusion matrices and the feature-space plots agree on why Aircraft is the harder dataset, and a full clean re-run reproduces every number bit for bit.

- The linear probe wins everywhere, and by a widening margin on Aircraft, up to +33.29 pp at full data, because averaging a class into one vector destroys fine-grained signal that a trained boundary can still use.
- DINOv2 substantially outperforms ResNet-18 on Aircraft (67.55% versus 38.15% with the full-data probe). That gap was visible before either classifier existed, as a roughly 10× wider separability margin in the frozen features.
- Training is stable across all 27 probe runs. The one unusual pattern, late Aircraft checkpoints despite rising validation loss, was tested directly rather than left as a guess; tripling the epoch budget moved accuracy by at most 0.21 pp.
- Aircraft is hard for a specific reason: fine-grained, within-family visual similarity that a class-mean vector cannot capture and a trained linear layer only partially can.
- We kept two preprocessing decisions deliberately separate. The copyright banner was cropped, since it is a source-image artifact that is free to remove and was tested to have no measurable effect. The centre crop's roughly 41% width loss on Aircraft was left in place, because it belongs to the checkpoint's own preprocessing and changing it would break comparability with the frozen encoder.

These numbers are the reference points for the next two stages. Everything we add from here, whether a flow-matching layer trained alongside the classifier or one placed in front of a frozen one, is measured against the probe accuracies in Table 3.